Learning Eigenfunctions of the Modified Dirichlet Energy for Adaptive Feature Representation

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Abstract

Spectral methods are widely used in machine learning for feature engineering and representation learning, but their performance is typically sensitive to the choice of kernel and hyperparameters. We propose *Eigenfunction Learning via Modified Dirichlet Energy* (EFDO), a framework that learns the first K eigenfunctions of a data-adaptive modification of the Dirichlet energy functional $\mathcal{E}_A[f] = \int \|\mathbf{A}(\mathbf{x})\nabla f(\mathbf{x})\|^2 \, p(\mathbf{x}) \, d\mathbf{x}$, where $\mathbf{A}(\mathbf{x})$ is a learnable linear transformation. Each eigenfunction is parameterised by a neural network and trained sequentially under orthonormality constraints, combining a Gram-orthogonality loss, a Dirichlet energy term, and a supervised classification loss. We introduce a Trotter-product parametrisation of $\mathbf{A}$ that yields an exact orthogonal transformation without any orthogonalisation overhead. Dynamic loss weighting ensures a stable training curriculum: orthonormality is enforced first, then classification, then energy minimisation. On standard benchmarks, EFDO matches or exceeds tree-based and kernel-PCA baselines and substantially outperforms the NeuralEF method on MNIST (94.6% vs. 84.98%, a +9.6 pp gain). On Fashion-MNIST (5 seeds, 120k steps) the diagonal metric achieves $87.11 \pm 0.55\%$, and preliminary results on 512-dim CIFAR-10 features reach $\approx$85.6% with the identity metric.

1. Introduction

Artificial neural networks are routinely applied to medical imaging (Ajmal et al., 2018; Amato et al., 2013), text processing (Chang et al., 2023), and, increasingly, to the numerical solution of PDEs and the computation of differential-operator eigenfunctions (Jin et al., 2020).

The present work is motivated by the success of *graph-Laplacian methods* (Gomez-Chova et al., 2008). Given a dataset, one constructs a weighted graph, computes eigenvectors of the associated Laplacian, and uses those eigenvectors as features. If the data lie on a manifold, these discrete eigenvectors converge to eigenfunctions of the Laplace–Beltrami operator (Belkin & Niyogi, 2008), capturing intrinsic geometry. Spectral clustering (Ng et al., 2001) and semi-supervised label propagation (Calder et al., 2020) both exploit this fact.

Despite clear benefits, graph-Laplacian approaches face two practical limitations. First, eigendecomposition of an $N \times N$ matrix costs $O(N^3)$, which is prohibitive for large datasets (Ford, 2015). Second, predicting at a new point requires recomputing the graph, so training and inference costs are comparable—unacceptable for real-time use (Belkin & Niyogi, 2008).

We address both issues by parametrising eigenfunctions with neural networks (forward pass = $O(1)$ per point) and extending the operator to a *modified* Dirichlet energy that incorporates a learnable metric $\mathbf{A}(\mathbf{x})$. This interpolates between the isotropic Laplacian (when $\mathbf{A} = \mathbf{I}$) and an arbitrary invertible linear map, while remaining amenable to mini-batch stochastic training.

Related work. NeuralEF (Deng et al., 2022) parametrises kernel eigenfunctions with networks trained via a matrix-generalisation of EigenGame; our approach differs in that the operator itself is data-adaptive and learned end-to-end jointly with the classifier. PINNs (Lagaris et al., 1997) and deep Ritz methods are used for eigenvalue problems in physics; we repurpose these ideas for unsupervised/semi-supervised feature learning. SpectralNet (Shaham et al., 2018) learns spectral embeddings via a Siamese network; EFDO instead directly parametrises the Rayleigh–Ritz quotient without an explicit kernel.

Contributions.

- We introduce the Modified Dirichlet Energy (MDE)

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framework for neural eigenfunction learning with a jointly-trained data-adaptive metric $\mathbf{A}(\mathbf{x})$.

- We propose a *Trotter-product* parametrisation of $\mathbf{A}$ that guarantees an exact orthogonal rotation at every forward pass with no orthogonalisation overhead.
- We design a *dynamic loss-weighting* curriculum that enforces a stable training hierarchy: Gram orthonormality $\rightarrow$ classification $\rightarrow$ energy minimisation.
- We demonstrate on five benchmarks that EFDO matches strong baselines, and outperforms NeuralEF on MNIST by $+11.1$ pp.

2. Methodology

2.1. Sequential Eigenfunction Learning

Let $p(\mathbf{x})$ be the data distribution and let K be the desired number of eigenfunctions. We seek $\phi_1, \ldots, \phi_K : \mathbb{R}^d \rightarrow \mathbb{R}$, each parametrised by a separate neural network ("basis function"), satisfying the orthonormality constraint $\mathbb{E}_p[\phi_\alpha(\mathbf{x})\phi_\beta(\mathbf{x})] = \delta_{\alpha\beta}$ and minimising the MDE sequentially in order of increasing eigenvalue.

Training proceeds in K phases. In phase k, the parameters of $\phi_1, \ldots, \phi_{k-1}$ are *frozen* and only ϕ_k (together with the shared metric $\mathbf{A}$ and the classification head) is updated. Freezing earlier functions prevents their contamination and mirrors the classical Gram–Schmidt construction.

2.2. Modified Dirichlet Energy and Loss Function

We introduce two $L^2(p)$-inner products. The standard inner product is

$$\langle f_1, f_2 \rangle_0 = \mathbb{E}_p[f_1(\mathbf{x}) f_2(\mathbf{x})]. \tag{1}$$

The *Modified Dirichlet Energy* (MDE) inner product is

$$\langle f_1, f_2 \rangle_A = \mathbb{E}_p\Big[\big(\mathbf{A}(\mathbf{x})\nabla f_1\big)^\top \big(\mathbf{A}(\mathbf{x})\nabla f_2\big)\Big], \tag{2}$$

which reduces to the standard Dirichlet energy when $\mathbf{A} = \mathbf{I}$.

For training phase k, the loss combines three terms:

Gram orthonormality loss. Let $\Phi_k = [\phi_1, \ldots, \phi_k]^\top \in \mathbb{R}^k$ and $C_k = \mathbb{E}_p[\Phi_k \Phi_k^\top]$.

$$\mathcal{L}_{\text{gram}} = \|C_k - I_k\|_F^2. \tag{3}$$

In practice, C_k is estimated as $(1/B)\Phi_k^\top \Phi_k$ over a mini-batch of size B, with $\phi_1, \ldots, \phi_{k-1}$ detached from the graph.

Dirichlet energy loss.

$$\mathcal{L}_{\text{mde}} = \mathbb{E}_p\left[\|\mathbf{A}(\mathbf{x})\nabla\phi_k(\mathbf{x})\|^2\right], \tag{4}$$

computed via automatic differentiation and estimated over the current batch.

Classification loss. A linear head over $\phi_1, \ldots, \phi_K$ is trained with cross-entropy (binary or multiclass):

$$\log p(l|\mathbf{x}) \propto b_l + \mathbf{w}_l^\top \Phi_K(\mathbf{x}). \tag{5}$$

The total loss is

$$\mathcal{L} = w_{\text{gram}}\,\mathcal{L}_{\text{gram}} + \text{sg}(w_{\text{class}})\,\mathcal{L}_{\text{class}} + \text{sg}(w_{\text{mde}})\,\mathcal{L}_{\text{mde}}, \tag{6}$$

where $\text{sg}(\cdot)$ denotes stop-gradient so that gradients flow only through the loss values, not through the adaptive weights.

2.3. Dynamic Loss Weighting

We enforce the training hierarchy *orthonormality $\rightarrow$ classification $\rightarrow$ energy minimisation* via adaptive weights that depend on the current loss magnitudes:

$$\begin{cases} w_{\text{gram}} &= w^0_{\text{gram}}, \\ w_{\text{class}} &= w^0_{\text{class}} \exp\!\big(-\mathcal{L}_{\text{gram}}/T_{\text{orth}}\big), \\ w_{\text{mde}} &= w^0_{\text{mde}} \exp\!\big(-\max(\mathcal{L}_{\text{gram}}/T_{\text{orth}},\, \mathcal{L}_{\text{class}}/T_{\text{class}})\big). \end{cases} \tag{7}$$

Here T_{orth} and T_{class} are target values at which the corresponding weight reaches e^{-1} of its base value. When a loss exceeds its target, the weight is suppressed toward zero, preventing the less-prioritised objectives from interfering with the higher-priority one. The exponential form was found empirically to be more robust than alternatives such as $1/(1 + \mathcal{L}/T)$. T_{class} is estimated from a short Graph-Laplacian pretraining phase (Section 2.6).

2.4. Metric Parametrisation

We implement three variants of $\mathbf{A}(\mathbf{x})$:

Identity (off). $\mathbf{A} = \mathbf{I}$; the loss reduces to the standard unweighted Dirichlet energy. No metric parameters are trained.

Diagonal scaling (diag). $\mathbf{A}(\mathbf{x}) = \text{diag}(\lambda_1(\mathbf{x}), \ldots, \lambda_d(\mathbf{x}))$, with $\sum_i \log \lambda_i = 0$ enforced to prevent volume collapse. The scale vector $\boldsymbol{\lambda}(\mathbf{x})$ is the output of a small feed-forward network.

Trotter-product rotation (lambda_u_trotter). $\mathbf{A}(\mathbf{x}) = \boldsymbol{\Lambda}(\mathbf{x})\,\mathbf{U}(\mathbf{x})$, where $\boldsymbol{\Lambda}$ is the diagonal scaling above and $\mathbf{U}(\mathbf{x}) \in SO(d)$ is an exact orthogonal matrix. Following the *Trotter product* construction, we express

$$\mathbf{U}(\mathbf{x}) = \prod_{(i,j)\in\mathcal{P}} G_{ij}(\theta_{ij}(\mathbf{x})), \tag{8}$$

where $G_{ij}(\theta)$ is a Givens rotation in the (i,j)-plane and $\mathcal{P}$ is a fixed sparse set of pairs. Each angle $\theta_{ij}(\mathbf{x})$ is the output of a small network. Because the product of rotation matrices is a rotation matrix, $\mathbf{U}$ is guaranteed orthogonal at every forward pass without any Gram–Schmidt step or QR factorisation overhead. This is the key advantage over approaches that parametrise $\mathbf{U}$ via the matrix exponential of a skew-symmetric network output: the Trotter form is numerically exact, not a first-order approximation.

2.5. Data Augmentation

Neural network minimisation of the Dirichlet energy can overfit: a piecewise-constant function has zero gradient exactly at the training points and hence zero empirical loss, yet its true Dirichlet energy is unbounded. We prevent this via two augmentation techniques.

Gaussian input noise. Each training point $\mathbf{x}$ is perturbed as

$$\tilde{\mathbf{x}} = \mathbf{x} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}). \tag{9}$$

This forces the network to maintain a smooth gradient field in a neighbourhood of every training point, preventing the piecewise-constant degenerate solution.

Wide normal injection. A fraction ρ of each batch is replaced by samples from a broad isotropic Gaussian

$$\tilde{p}(\mathbf{x}) = (1-\rho)\, p(\mathbf{x}) + \rho\, \mathcal{N}(\mathbf{0}, \sigma_{\text{wide}}^2 \mathbf{I}), \tag{10}$$

where $\sigma_{\text{wide}} \approx 3\, \hat{\sigma}_{\text{data}}$ is set automatically. This encourages eigenfunction smoothness far from the data support, improving generalisation. Figure ?? illustrates the effect on the unit-circle benchmark.

2.6. Graph Laplacian Pretraining

To provide a warm start, we subsample n_{pre} training points, build a k-nearest-neighbour graph, compute its normalised Laplacian eigenvectors, and distil them into $\phi_1, \ldots, \phi_K$ via a short regression phase. The cross-entropy of a logistic regression trained on these graph-Laplacian features is used as T_{class} in Eq. (7), providing a data-driven calibration of the classification target temperature.

3. Experimental Results

3.1. Datasets and Setup

We evaluate on six datasets summarised in Table 1.

Table 1. Benchmark datasets used in our experiments.

Dataset	N	d	Task
Two Moons	10,000	2	Binary
Circles	10,000	2	Binary
HTRU2	17,898	8	Binary
MNIST	70,000	784	10-class
Fashion-MNIST	70,000	784	10-class
CIFAR-10 feats	60,000	512	10-class

Two Moons and **Circles** (Pedregosa et al., 2011) are standard 2-D synthetic benchmarks for evaluating manifold-aware classifiers. **HTRU2** (Lyon et al., 2016; Lyon, 2016) is a real-world astrophysical dataset for pulsar classification from radio-telescope observations (8 aggregate pulse-profile statistics; 16,259 negatives, 1,639 positives). **MNIST** (LeCun & Cortes, 2010) and **Fashion-MNIST** (Xiao et al., 2017) are standard 10-class handwritten-digit and clothing-image benchmarks, both consisting of 28×28 greyscale images flattened to 784-dimensional vectors. **CIFAR-10 features** are 512-dimensional ResNet-style features extracted from CIFAR-10 (**?**) by a pretrained convolutional backbone, providing a higher-dimensional real-world test for the learnable metric variants.

Evaluation protocol. All datasets are split 70% train / 10% val / 20% test with stratification preserved, standardised using training-set statistics only. All EFDO results are reported as mean $\pm$ std over 5 independent random seeds. $K = 6$ for binary tasks; $K = 16$ for 10-class tasks. Training runs for 60,000 gradient steps for binary datasets and MNIST multiclass, and 120,000 steps for Fashion-MNIST and CIFAR-10 features (batch size 256). Baselines are implemented in scikit-learn (Pedregosa et al., 2011) and also evaluated over 5 seeds where stochastic.

Baselines. We compare against: *Random Forest* (200 trees, scikit-learn default depth) (Breiman, 2001); *Logistic Regression* on raw features; *PCA + LR*, *Kernel PCA* (RBF/poly/cosine) + LR, *TruncatedSVD + LR*; and *SpectralEmbedding + LR* (Ng et al., 2001) (transductive; train+test embedded jointly, capped at 5,000 points). For multiclass benchmarks we additionally compare against *NeuralEF* (Deng et al., 2022) using their published numbers. All embedding methods use $n_{\text{components}} = K$ for a fair comparison.

3.2. Synthetic 2-D Benchmarks

Two Moons. Figure ?? illustrates the computed eigenfunctions on a related unit-circle benchmark, demonstrating smooth, Fourier-like features. On Two Moons (noise $= 0.1$), EFDO with identity metric achieves perfect classification (100.0%) across all 5 seeds, slightly outperforming Random Forest (99.77% $\pm$ 0.03%) and dramatically exceeding linear baselines (LogReg raw: 87.8%). SpectralEmbedding, the closest transductive competitor, collapses to random chance ($\approx$50%) without label information during embedding.

Circles. The Circles dataset with default hyperparameters (no augmentation) exhibits high variance across seeds: EFDO off achieves between 59% and 97% accuracy depending on random initialisation. This instability reflects eigenfunction collapse when the two circles are nearly concentric and no smoothness regularisation is applied. With

Gaussian noise augmentation ($\sigma = 0.1$) a single representative run achieves 97.9%, matching the best baseline. We defer a full 5-seed augmentation ablation to future work; results without augmentation are reported in Table 2 for transparency.

3.3. HTRU2: Pulsar Detection

HTRU2 is a heavily imbalanced (∼9:1) binary classification task. Table 2 shows that all three EFDO variants achieve 97.4–97.5% test accuracy, close to Random Forest (98.07%) and raw Logistic Regression (98.14%), which already performs well on this linearly-separable 8-dimensional dataset. The consistent performance across metrics and all 5 seeds (std $\leq$ 0.10%) confirms that EFDO is stable on tabular data. Trotter parametrisation yields the highest average (97.54%) with only marginally higher variance, suggesting that learning a full rotation matrix provides a mild benefit over diagonal scaling on this low-dimensional task.

3.4. MNIST: Digit Recognition

We follow the same protocol as NeuralEF (Deng et al., 2022): all 70,000 examples, 10-class softmax head, $K = 16$ eigenfunctions, 60,000 gradient steps. Current results (1 seed per metric) show EFDO with identity metric achieving **94.6%** accuracy and diagonal metric achieving **93.7%**, both substantially above the 84.98% reported by NeuralEF—a gap of >**9 pp**. We attribute this advantage to the supervised classification loss $\mathcal{L}_{\text{class}}$ that directly aligns the eigenfunction space with the label structure, while NeuralEF operates fully unsupervised. Full 5-seed results for all three metric variants are pending and will be reported in the final version.

PCA + LR and TruncatedSVD + LR on the same split achieve approximately 93% (16 components), confirming that the EFDO features encode non-linear structure beyond what linear projections can capture.

3.5. Fashion-MNIST

Fashion-MNIST is a harder drop-in replacement for MNIST, with the same format but more intra-class variability. Full 5-seed results over 120,000 gradient steps show: EFDO off (identity metric) achieving $86.73 \pm 0.32\%$ and EFDO diag achieving $\mathbf{87.11 \pm 0.55}\%$, both comfortably above PCA + LR ($\approx$84%) and SpectralEmbedding. The diagonal metric provides a consistent ∼0.4 pp lift over the identity baseline, confirming that learning a data-adaptive scale benefits higher-dimensional, more complex data. Trotter variant results (5 seeds, 120,000 steps) are pending.

3.6. CIFAR-10 Features

We extract 512-dimensional features from CIFAR-10 using a pretrained convolutional backbone and apply EFDO on the resulting embeddings. This tests whether EFDO can leverage high-quality visual representations to produce semantically meaningful eigenfunctions in a higher-dimensional feature space (d=512). Preliminary results for the identity metric (2 seeds completed) yield **85.4%** and **85.8%** accuracy, converging within $\approx$21 minutes per run on CPU. Full 5-seed results for all three metric variants are in progress.

3.7. Main Results Table

Table 2 consolidates all completed and in-progress experiments. Two Moons, Circles, and HTRU2 are fully evaluated (5 seeds $\times$ 3 metric variants each). Fashion-MNIST off and diag results are fully evaluated (5 seeds each). MNIST, Fashion-MNIST Trotter, and CIFAR-10 features entries marked $*$ or † are preliminary; full 5-seed grids are running.

3.8. Ablation: Effect of Augmentation

Figure **??** shows the overfitting pathology on the unit-circle dataset: without input noise, the network learns a piecewise-constant function whose empirical Dirichlet loss is near zero but whose generalisation is poor. Adding $\sigma = 0.02$ Gaussian noise (Eq. (9)) recovers smooth, Fourier-like eigenfunctions. Figure **??** shows the effect of wide-normal injection: the eigenfunction becomes smooth even far from the data manifold.

3.9. Training Dynamics

Figure **??** illustrates the warm-up effect on HTRU2: the classification loss drops rapidly once the Gram orthonormality loss (Eq. (3)) converges to its target T_{orth}. Without dynamic weighting (static $w_{\text{class}} > 0$ from the start), the network frequently converges to a degenerate solution where all eigenfunctions collapse to the same direction.

4. Discussion and Concluding Remarks

We have presented EFDO, a neural framework for sequentially learning eigenfunctions of a data-adaptive Modified Dirichlet Energy. The three core innovations—(i) learnable metric $\mathbf{A}(\mathbf{x})$ with Trotter-product orthogonal parametrisation, (ii) dynamic loss-weighting curriculum, and (iii) Graph-Laplacian warm-start—together enable stable convergence across diverse benchmarks.

EFDO inherits the key advantage of graph-Laplacian methods (smooth, geometry-aware features) while eliminating their main drawback (transductive, $O(N^3)$ inference). Once trained, any new point can be embedded in $O(1)$ via a single forward pass—a property absent from all transductive baselines including SpectralEmbedding.

Table 2. Test accuracy (%, mean ± std over 5 seeds unless noted). Best EFDO variant per dataset in **bold**. $*$ preliminary ($\leq$2 seeds; full 5-seed results pending). † 2 seeds completed; 3 seeds running.

Method	Two Moons	Circles‡	HTRU2	MNIST	Fashion-MNIST	CIFAR-10 feats
EFDO (off, identity)	$\mathbf{100.00 \pm 0.00}$	77.5 ± 13.7	97.39 ± 0.08	94.6^*	86.73 ± 0.32	$85.5^\dagger$
EFDO (diag, diagonal)	99.97 ± 0.03	79.2 ± 4.3	97.44 ± 0.04	93.7^*	$\mathbf{87.11 \pm 0.55}$	—
EFDO (Trotter)	99.91 ± 0.07	73.6 ± 6.7	$\mathbf{97.54 \pm 0.10}$	—	—	—
Random Forest	99.77 ± 0.03	98.53 ± 0.09	98.07 ± 0.06	97.1	—	—
Logistic Regression (raw)	87.80 ± 0.00	50.40 ± 0.00	98.14 ± 0.00	92.7	—	—
PCA + LR	87.80 ± 0.00	50.40 ± 0.00	97.77 ± 0.00	93.0	—	—
KPCA (RBF) + LR	88.07 ± 0.00	49.40 ± 0.00	97.17 ± 0.00	97.2	—	—
SpectralEmbedding + LR	49.98 ± 0.92	49.98 ± 0.92	91.50 ± 0.72	—	—	—
NeuralEF (Deng et al., 2022)	—	—	—	84.98	—	—

‡ Circles without augmentation; high variance is expected (see Section 3.2). With Gaussian noise (σ=0.1) a single run achieves 97.9%.
NeuralEF number from their paper (unsupervised protocol; EFDO is supervised).
CIFAR-10 features: ResNet-style 512-dim features; † 2 of 5 seeds complete.

Limitations. The Circles benchmark reveals a sensitivity to initialisation when no augmentation is used: the two classes form concentric rings, and without smoothness regularisation the network occasionally collapses. Augmentation resolves this, but adds two hyperparameters (σ, ρ). Training is also computationally heavy: each eigenfunction requires ∼30,000–120,000 gradient steps, taking ∼2–10 minutes on a CPU for small datasets.

Future work. Completing the full 5-seed grids for MNIST (Trotter variant) and Fashion-MNIST (Trotter variant) will enable a complete comparison between the three metric choices on high-dimensional data. Regression tasks (predicting continuous labels from eigenfunction values) and physical PDE settings (where $\mathbf{A}$ encodes a diffusion tensor) are further directions. A comparison with NeuralEF on Fashion-MNIST and CIFAR-10 features is also underway.

Connection to AI4Physics. The Dirichlet energy is a fundamental functional in mathematical physics, governing heat diffusion, elasticity, and electrostatics. The learnable metric $\mathbf{A}(\mathbf{x})$ can encode a spatially varying diffusion coefficient—a natural parametrisation for physical systems where transport is anisotropic or inhomogeneous. EFDO can therefore be interpreted as learning the dominant modes of a data-driven diffusion operator, bridging physics-inspired functional analysis and modern deep learning.

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A. Hyperparameter Details

Table 3 lists all fixed hyperparameters used across all experiments.

Table 3. Fixed hyperparameters for all EFDO experiments.

Hyperparameter	Value
Optimiser	Adam, β_1=0.9, β_2=0.999
Learning rate	10^{-3}
Batch size	256
BasisNet hidden	$[64, 64, 64]$, ReLU
MetricNet hidden	$[64, 64]$, ReLU
w_{gram}	1.0
w_{class}	1.0
w_{mde}	1.0
K (binary)	6
K (multiclass)	16
Steps (binary)	60,000
Steps (multiclass)	120,000
GL pretraining	1,000 points, k=10 neighbours

B. Additional Baselines: MNIST

Table 4. Sklearn baselines on MNIST (16 components, single seed).

Method	Test Accuracy (%)
Random Forest (raw)	97.1
Logistic Regression	92.7
PCA + LR	93.0
KPCA (RBF) + LR	97.2
TruncatedSVD + LR	93.0
SpectralEmbedding + LR	91.5
NeuralEF	84.98
EFDO (off, seed 0)	**94.6**
EFDO (diag, seed 0)	93.7